Nero Ziyu Li

For solo papers I use the name Ziyu Neroli.

I love the theorem that, in form, appears as elegant and simple as a blossom of neroli; yet its proof, when tasted, reveals the bitterness of an orange.

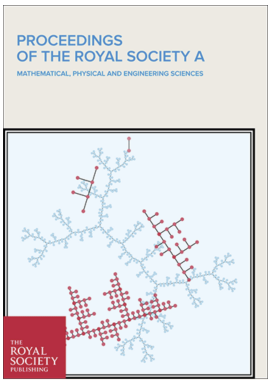
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Personal page
sites.google.com/view/neroziyuli

Google Scholar
scholar.google.com/citations?user=fHiwGoEAAAAJ

Interests

- Math
- Physics
- Machine learning/AI



Education

Ph.D. in Mathematics (Now)

Imperial College London, United Kingdom
*with **CDT Random Systems** at the University of Oxford*
supervised by Professor Martin Rasmussen and Professor Jeroen Lamb

Master by Research

University of New South Wales (UNSW Sydney), Australia
supervised by Professor Thomas Britz

B.Sc. in Mathematics

Ningbo University, China
supervised by Professor Lifeng Xi

Awards and Funding

2019 **New World Mathematics Awards**
Gold Medal in the 2017 competition, presented by Fields Medallist Shing-Tung Yau (only two Gold Medals and three Silver Medals were awarded worldwide across the 2017–2019 competitions).

2020 **Best Speaker**, pure mathematics session, UNSW Annual Postgraduate Conference.

2021 **Heilbronn Scholarship**
PhD funded by Heilbronn Institute for Mathematical Research (for UK residents) and Imperial College London (international fee difference) (EP/V521917/1).

2021 **CDT Random Systems**
Also funded by the EPSRC Centre for Doctoral Training (CDT) in Mathematics of Random Systems (EP/S023925/1) at University of Oxford.

2023 **Unique People's Choice Winner**, Imperial Mathematics Poster Competition.

2026 **Third Prize**, Best Poster Competition, 77th British Mathematical Colloquium, Cardiff University.

Publications and Preprints

Mathematics

Solutions of $x^\alpha + y^\alpha = 1$ in λ-Cantor sets, arXiv:2610.29793, submitted, 2026. (36 pages, with Kan Jiang & Lifeng Xi)
The solutions exhibit the Hausdorff phase transition, solving the open question left in [J. London Math. Soc. 113 (2026), e70408].

Iterated Graph Systems (III): Alexander-Orbach conjecture on hierarchical lattices, arXiv:2610.16875, 2026. (36 pages)
Outside a narrow region in dimension space, we prove that the Alexander–Orbach conjecture fails on every hierarchical lattice under consideration.

Iterated Graph Systems (II): Bernoulli percolation and universality classes on hierarchical lattices, arXiv:2610.10469, 2026. (45 pages)
No finite family of substitution-multiplicative dimensions completely determines universality, providing a rigorous proof of the failure of dimension-based universality discussed by Hauser and Saxena [Phys. Rev. B 34 (1986), 8193–8195].

Iterated Graph Systems (I): random walks and diffusion limits, arXiv:2603.13798, 2026. (50 pages)
Unifies the spectral dimension of EIGS and solves the open problem on DHL percolation cluster posed in [Commun. Math. Phys. (2010) 295, 29–69].

Reducible Iterated Graph Systems: multiscale-freeness and multifractals, Nonlinearity, 2026. (26 pages, with Thomas Britz, Frank Hu)
Completes the fundamental theory of EIGS by investigating the reducible regime.

Featured on the issue cover: Fractal dimensions for iterated graph systems, Proceedings of the Royal Society A, 2024. (28 pages)
See the summary news in Chinese by the Royal Society
Establishes the iterated graph systems.

On the scale-freeness of random colored substitution networks, Proceedings of the American Mathematical Society, 2024. (13 pages, with Thomas Britz)
Defines the degree dimension for graphs.

Physics

Z. Neroli Moiré Pattern of Rotated Sierpiński Gaskets, arXiv:2609.30235, 2026.

Z. Li, J. Yao, Q. Wang, Fractality of multiple colored substitution networks, Physica A, 2019.

GitHub

github.com/Nero-17

Four-Agent Research

This reusable four-agent workflow coordinates research planning, conjecture generation, proof development and independent review, while adapting its research mode and reasoning effort and archiving cross-session LaTeX reports.

Z. Li, Z. Yu, L. Xi, Scale-free effect of substitution networks, Physica A, 2018.

Machine Learning/AI

Nero Z. Li, et al, Fractal Graph Contrastive Learning, arXiv:2505.11356, 2025.

Talks

2026 54th Probability Summer School Saint-Flour; Berlin Summer School on Applied Stochastic Analysis, TU Berlin; 77th British Mathematical Colloquium, Cardiff.

2025 Junior Dynamics Seminar, Imperial College London.

2024 Geometry and Fractals under the Midnight Sun, University of Oulu; Heilbronn PhD Student Research Day, Bristol.

2023 Dynamical Systems Seminar, The Open University.

2022 Mathematics of Random Systems CDT Spring Retreat, Missenden Abbey.

2020 UNSW Annual Postgraduate Conference.

2019 42nd Australasian Conference on Combinatorial Mathematics and Combinatorial Computing, UNSW Sydney.

Conference Participation

2026 54th Probability Summer School Saint-Flour; Berlin Summer School on Applied Stochastic Analysis, TU Berlin; 77th British Mathematical Colloquium, Cardiff.

2025 INI/IMU Mathematical Colloquia, Isaac Newton Institute for Mathematical Sciences, University of Cambridge.

2024 Geometry and Fractals under the Midnight Sun, University of Oulu; Heilbronn PhD Student Research Day, Bristol.

2023 Workshop in honour of Peter Baxendale, University of Warwick; CDT Summer School, Kyoto University, Japan; CHAOS Workshop, Nesin Mathematics Village, Turkey.

2022 ICM 2022 Overlay, Imperial College London; Heilbronn Summer Workshop at ICMS, Edinburgh; Durham Symposium: Stochastic Dynamics, Nonlinear Probability, and Ergodicity, Durham University; UK-Japan Winter School; Mathematics of Random Systems CDT Spring Retreat, Missenden Abbey.

2020 UNSW Annual Postgraduate Conference.

2019 42nd Australasian Conference on Combinatorial Mathematics and Combinatorial Computing, UNSW Sydney; International Congress of Chinese Mathematicians and NWMA Ceremony (invited).

2017 China Fractal and Dynamical System Academic Conference (invited).

Teaching and Supervision

Imperial College London

2026 **Supervision of Master's thesis** related to bounded-noise Hopfield networks.

2023–24 **Graduate Teaching Assistant**:
Autumn: **MATH40003 Linear Algebra and Groups**.
Spring: **MATH40003 Linear Algebra and Groups**; **MATH50007 Network Science**.
Summer: **LaTeX workshops**.

2022–23 **Graduate Teaching Assistant**:
Autumn: **MATH50003 Linear Algebra and Numerical Analysis**; **MATH60008 Dynamical Systems**.
Spring: **MATH50003 Linear Algebra and Numerical Analysis**.

Experiences

Reviews for journals Physica A, FILOMAT and the Australasian Journal of Combinatorics.

Life member of the Combinatorial Mathematics Society of Australasia; Member of Australian Mathematical Society; Member of London Mathematical Society.

Master ball tier (top 2000 globally) for double battle in Pokémon Sword and Shield, with only ghost Pokémons; Gold-medal top 100 with Mengqi and Baili Xuance in Honor of Kings;
Band Leader, K-ON (2014–2016); Bamboo flute/percussion, CMS UNSW (2018–2021); Keytar, Melon Band (2022–2024); Music busking in London; Reach the standard of the Chinese national third-grade athlete (200m, 24.60s).